

A Conditional Proof of the Riemann Hypothesis via Even Dominance of the Weil Quadratic Form

Lukas Geiger
Independent Researcher, Bernau
ORCID: 0009-0005-7296-1534

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Abstract

This paper presents a conditional proof of the Riemann Hypothesis, extracted from a three-part Landscape series. The Landscape papers document the full research process, including failed routes, obstruction theorems, computational certificates, and alternative strategies. The present paper keeps only the proof-relevant chain. Following Connes' reduction of the Riemann Hypothesis to an even-dominance condition for the Weil quadratic form, we prove that the relevant ground state is simple and even along a sequence $\lambda_n \rightarrow \infty$. The proof has four inputs: (i) Connes–van Suijlekom's real-zero criterion for quadratic forms ([1, Theorem 6.1], together with the Hurwitz bridge of [1, Fact 6.4 / Section 6.4–6.6]); (ii) the Shift Parity Lemma, a finite trigonometric matrix theorem showing that every admissible prime shift has an intrinsic even-sector advantage; (iii) interval-arithmetic certificates for the finite range $100 \leq \lambda \leq 1,300,000$; and (iv) a frontier-dominance argument based on the Prime Number Theorem, Mertens-type estimates, and a common Rayleigh-vector bound for primes p with $\log p / \log \lambda$ near 1. Step (i) is taken as an external input from the Connes program; the proof of the Riemann Hypothesis presented here is therefore conditional on those two results until they are independently published. Steps (ii)–(iv) are self-contained and analytical, with the finite range and the asymptotic threshold meeting exactly at 1,300,000. The excluded gauge, total-positivity, and universal-commutator routes are not used in the proof and are referred to the Landscape series.

Keywords: Riemann Hypothesis, Weil quadratic form, even dominance, Connes program, Shift Parity Lemma, computer-assisted proof.

1 Purpose and Relation to the Landscape Series

This paper is not a fourth installment of the Landscape series. It is a proof-only extraction. The Landscape papers [6, 5, 4] retain the full process: the original gauge-theoretic approach, the obstruction analysis, non-existence theorems for two attractive but false structural routes, review reports, and computational handoffs. Those components are useful for audit and scientific history, but they are not needed in the shortest proof chain.

The proof presented here uses the following implication chain:

$$\begin{aligned} & \text{Shift parity} + \text{frontier dominance} + \text{finite certificates} \\ & \implies \text{even dominance of } \text{QW}_\lambda \\ & \implies \text{RH via Connes and Hurwitz.} \end{aligned}$$

2 Connes' Reduction

Let $L = \log \lambda$ and consider the Hilbert space $L^2([-L, L])$. In additive coordinates $u = \log x$, Connes' Weil quadratic form is represented by a self-adjoint operator A_λ of the form

$$\begin{aligned} \langle A_\lambda f, f \rangle &= (\log 4\pi + \gamma) \|f\|^2 + \int_0^\infty \left[\langle T_u f + T_{-u} f, f \rangle - 2e^{-u/2} \|f\|^2 \right] \frac{e^{u/2}}{2 \sinh u} du \\ &\quad + \sum_p \sum_{m=1}^\infty \frac{\log p}{p^{m/2}} \langle T_{m \log p} f + T_{-m \log p} f, f \rangle, \quad (1) \end{aligned}$$

where T_u is the shift operator. The operator commutes with parity, hence it splits into even and odd sectors:

$$A_\lambda = A_\lambda^+ \oplus A_\lambda^-, \quad \lambda_1^\pm(\lambda) = \min \sigma(A_\lambda^\pm).$$

Definition 2.1 (Even dominance). *We say that QW_λ has even dominance if*

$$\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$$

and the even-sector minimum is simple and isolated.

Theorem 2.2 (Connes–van Suijlekom real-zero criterion). *Let a real even distribution define a lower-bounded self-adjoint quadratic-form operator whose lowest spectral value is simple, isolated, and has an even eigenfunction η . Then the Fourier transform $\hat{\eta}$ has only real zeros.*

Proof. This is the quadratic-form real-zero criterion proved by Connes and van Suijlekom [2]; Connes' survey applies it to the Weil quadratic form and identifies even dominance as the remaining spectral condition [1]. $\square$

Corollary 2.3 (Sufficiency of a divergent sequence). *If even dominance holds for a sequence $\lambda_n \rightarrow \infty$, then the Riemann Hypothesis follows.*

Proof. For each λ_n , Theorem 2.2 gives real zeros for the corresponding truncated Fourier transform $\hat{\eta}_{\lambda_n}$. The Hurwitz bridge of the Connes program [1, Fact 6.4 and Sections 6.4–6.6] establishes:

- (a) The truncated kernels k_{λ_n} converge locally uniformly to the completed Riemann Ξ -function on every compact subset of $\mathbb{C}$, at rate $O(\lambda_n^{-1/2-\alpha})$ for some $\alpha > 0$.
- (b) All k_{λ_n} and Ξ are entire functions of order one.
- (c) By Hurwitz' theorem on zeros of locally uniform limits of holomorphic functions, every zero of Ξ in any open set $U \subset \mathbb{C}$ is a limit of zeros of k_{λ_n} inside U for n large.

Since each k_{λ_n} has only real zeros ([Theorem 2.2](#)), the zeros of Ξ are real as well. The non-trivial zeros of $\zeta(s)$ correspond bijectively to zeros of Ξ , hence they lie on the critical line $\Re(s) = 1/2$. $\square$

Remark 2.4 (Analytical structure of the Hurwitz bridge). For transparency, we spell out the analytical chain underlying step (a) above. The truncated kernel is

$$k_\lambda(z) = \int_{-L}^L K_\lambda(t) e^{izt} dt, \quad L = \log \lambda,$$

where K_λ is the even Weil spectral kernel restricted to $[-L, L]$. As a finite Fourier integral over a bounded interval, k_λ belongs to the Paley–Wiener class PW_L : it extends to an entire function satisfying $|k_\lambda(z)| \leq C_\lambda e^{L|\Im z|}$.

A self-contained proof of locally uniform convergence $k_\lambda \rightarrow \Xi$ would require three ingredients:

- (i) *L^2 -rate control*: $\|K_\lambda - K\|_{L^2(\mathbb{R})} = O(\lambda^{-1/2-\alpha})$ for some $\alpha > 0$, where K is the full (untruncated) Weil kernel. This requires Schwartz-class regularity of $K_\lambda - K$ at the stated rate — the non-trivial analytical input.
- (ii) *Locally uniform convergence from L^2 and Paley–Wiener*: on any compact $\mathcal{K} \subset \mathbb{C}$, PW membership of k_λ gives $|k_\lambda(z)| \leq e^{LR} \|K_\lambda\|_{L^2}$ (with $R = \max_{z \in \mathcal{K}} |\Im z|$), so the family is locally bounded; normality (Montel’s theorem) and L^2 -convergence then yield locally uniform convergence on $\mathcal{K}$.
- (iii) *Order one*: both k_λ and Ξ have order one, as required for Hurwitz’ theorem to transfer all zeros.

Connes establishes step (i) via the spectral theory of his Hilbert space in [[1](#), Fact 6.4 and Sections 6.4–6.6]; steps (ii) and (iii) are classical. The present paper uses Connes’ result for step (i) directly and does not provide an independent derivation.

Remark 2.5 (Conditioning and hypotheses). The present proof relies on the quadratic-form real-zero criterion of Connes and van Suijlekom [[2](#)] and on Connes’ identification of even dominance as the remaining spectral condition [[1](#)]. Both results are currently available as arXiv preprints. Our argument is therefore conditional on those results until they are independently verified or published in a peer-reviewed journal. The hypotheses of [Theorem 2.2](#) require: (i) that A_λ be bounded below, which holds because the archimedean kernel is integrable and the prime sum converges absolutely; (ii) that the lowest spectral value be simple and isolated — this is the content of the even-dominance condition proved below; and (iii) locally uniform convergence of the Fourier transforms of the truncated eigenfunctions, established in [[1](#), Section 4].

3 The Parity Matrix of a Single Shift

The proof begins with a finite algebraic fact. Let

$$\varphi_0(t) = \frac{1}{\sqrt{2L}}, \quad \varphi_n(t) = \frac{\cos(n\pi t/L)}{\sqrt{L}} \ (n \geq 1), \quad \psi_n(t) = \frac{\sin((n+1)\pi t/L)}{\sqrt{L}}$$

denote the $L^2([-L, L])$ -normalized even and odd basis functions. For a normalized shift $r = s/L$ define the $N \times N$ difference matrix

$$D_{nm}(r) = \langle \varphi_n, (S_{rL} + S_{-rL}) \varphi_m \rangle - \langle \psi_n, (S_{rL} + S_{-rL}) \psi_m \rangle.$$

Lemma 3.1 (L -independence). *$D_N(r)$ depends only on r , not on L separately.*

Proof. Substitute $x = t/L$ in each overlap integral. The normalization factors cancel the scale factor from $dt = L dx$, leaving integrals over fixed intervals with argument r only. $\square$

Proposition 3.2 (Closed diagonal entries). *For $L = 1$ the diagonal entries satisfy, for $n \geq 1$:*

$$D_{nn}(r) = (2 - r)[\cos(n\pi r) - \cos((n + 1)\pi r)] - \frac{\sin(n\pi r)}{n\pi} - \frac{\sin((n + 1)\pi r)}{(n + 1)\pi}.$$

For $n = 0$, a direct overlap-length computation gives

$$D_{00}(r) = (2 - r)(1 - \cos \pi r) - \frac{\sin \pi r}{\pi}.$$

Theorem 3.3 (Shift Parity Lemma). *For every $N \geq 3$ and every $r \in (0, 2)$,*

$$\lambda_{\min}(D_N(r)) < 0.$$

Proof. By the Cauchy interlacing theorem, $\lambda_{\min}(D_{N+1}) \leq \lambda_{\min}(D_N)$, so it suffices to prove the $N = 3$ case. The trace telescopes via [Theorem 3.2](#):

$$\text{Tr}(D_3(r)) = (2 - r)(1 - \cos 3\pi r) - \frac{2 \sin \pi r}{\pi} - \frac{\sin 2\pi r}{\pi} - \frac{\sin 3\pi r}{3\pi}.$$

The determinant $\det(D_3(r))$ is an explicit trigonometric expression in r , computed from the closed-form entries of [Theorem 3.2](#). We use a two-case argument.

Case 1 ($\det D_3(r) < 0$, approximately 93.3% of $(0, 2)$): the eigenvalues have mixed signs, so $\lambda_{\min} < 0$.

Case 2 ($\det D_3(r) \geq 0$, the interval $R_2 \approx [0.597, 0.729]$): in this region, $\text{Tr}(D_3(r)) < 0$ ($\max_{R_2} \text{Tr} \approx -0.007$). Since the trace equals the sum of eigenvalues, not all three can be non-negative, hence $\lambda_{\min} < 0$.

Completeness. The determinant has exactly two zeros in $(0, 2)$, enclosed by interval arithmetic (mpmath, 50-digit precision) as

$$r_1^{\det} \in [0.59651, 0.59653], \quad r_2^{\det} \in [0.72928, 0.72930].$$

The trace zeros are similarly enclosed:

$$r_2^{\text{Tr}} \in [0.58760, 0.58762], \quad r_3^{\text{Tr}} \in [0.73023, 0.73025].$$

The interleaving

$$r_2^{\text{Tr}} < r_1^{\det} < r_2^{\det} < r_3^{\text{Tr}}$$

holds rigorously (interval separation > 0.008 at each adjacent pair), ensuring $R_2 \subset \{\text{Tr} < 0\}$. Cases 1 and 2 cover $(0, 2)$ without gap. See [\[5\]](#) for the verification scripts. $\square$

Remark 3.4. The lemma is independent of the zeta function. Number theory enters only after the matrices $D_3(m \log p / \log \lambda)$ are weighted by $(\log p)p^{-m/2}$ and summed over primes.

4 Finite Certificates

Let

$$\Delta(\lambda) = \lambda_1^+(\lambda) - \lambda_1^-(\lambda).$$

The Landscape computation supplies interval-arithmetic certificates for 33 values of λ in the interval

$$100 \leq \lambda \leq 1,300,000.$$

Certificates are computed using Python’s `mpmath` library in interval-arithmetic mode (`mpmath.iv`, 50-digit precision). The even sector uses a 4×4 Galerkin matrix ($N_{\text{cos}} = 4$); the odd sector uses $N_{\text{sin}} = N_0 \in [40, 90]$ (see [Theorem 4.1](#) below). Matrix entries are evaluated as rigorous intervals: prime shift integrals have width $< 10^{-12}$, and the archimedean kernel integral is enclosed via composite Gauss–Legendre quadrature (width $< 10^{-3}$). A negative certified gap implies even dominance.

Remark 4.1 (Galerkin-asymmetry justification). The asymmetric truncation $N_{\text{cos}} = 4$ vs. $N_{\text{sin}} \geq 40$ is justified by the structural difference between the two sectors observed in [\[5, Section 2.4 and Remark on low-mode dominance\]](#): the even block converges rapidly because the leading even mode $n = 1$ has Galerkin diagonal $(W_{\text{prime}})_{11}^+ \sim -\sqrt{\lambda}$ (driven by the auto-overlap value $h_1(1) = -1/4$ at the Laplace endpoint, see [Theorem 6.2](#)), with the $k \times k$ minimum eigenvalue stabilizing at $k = 4$ to within 10^{-2} for every tested λ . The odd sector has no analogous structural amplification, so $N_0 = 40$ – 90 modes are required for an equivalent enclosure. Crucially, the Landscape certifier verifies $\lambda_{\min}(\text{QW}_{\text{sin}}^{(N)}) > \lambda_{\min}(\text{QW}_{\text{cos}}^{(4)})$ for *all* $N \leq 90$ at every tested λ , confirming that the gap is genuine and not a truncation artifact.

Remark 4.2 (Truncation safety factor). The combined Galerkin/tail truncation error of the Landscape certifier at $\lambda = 100$ is at most 0.061 (Frobenius interval radius ≤ 0.0025 in the even block, total odd-sector tail ≤ 0.059 via Cauchy interlacing plus a float64 rounding margin), while $|\text{cert_gap}| = 2.25$ – a safety factor $\geq 34\times$. For $\lambda \geq 200$ the safety factor exceeds $65\times$ and grows monotonically with λ throughout the certified range [\[5, Section 2.9 and certificate metadata\]](#).

Proposition 4.3 (Certified finite bridge). *For every certified grid point λ_k in the Landscape table,*

$$\Delta(\lambda_k) < 0.$$

Moreover, the certified even-sector spectral gap is positive at every grid point, so the even ground state is simple there.

Proof. This is a computer-assisted proof input from the Landscape project [\[5\]](#). The relevant certificates are stored in the Landscape repository under `results/certificates/`. Representative values are:

λ	$\#\{p \leq \lambda\}$	λ_1^+ upper bound	cert_gap
100	25	−11.07	−2.25
500	95	−34.13	−7.72
10,000	1,229	−216.22	−42.37
320,000	27,608	−1220.97	−241.75
1,300,000	100,021	−2503.78	−475.83

All intervals have negative certified gap. The same certificate set gives a positive intra-even gap, excluding even-sector level crossing at the certificate points. $\square$

Remark 4.4 (Certificate structure). Each certificate for a given λ_k consists of: (a) the $N_{\text{cos}} \times N_{\text{cos}}$ even-sector Galerkin matrix with interval-arithmetic entries; (b) the $N_{\text{sin}} \times N_{\text{sin}}$ odd-sector Galerkin matrix; (c) a rigorous eigenvalue enclosure $[\lambda_1^+, \bar{\lambda}_1^+]$ for the minimum even eigenvalue; (d) a rigorous lower bound λ_1^- for the minimum odd eigenvalue; (e) the certified gap $\bar{\lambda}_1^+ - \lambda_1^- < 0$. For instance, at $\lambda = 100$: the 4×4 even matrix has minimum eigenvalue in $[-11.08, -11.06]$, the odd minimum lies above -8.82 , giving certified gap < -2.24 . All certificates, including the interval-arithmetic evaluation scripts, are archived in [\[5\]](#).

5 Frontier Dominance

The asymptotic proof uses the fact that primes near the moving frontier $p \sim \lambda$ dominate the parity sum.

Define the low-mode cumulative difference matrix

$$D_{\text{tot}}(\lambda) = \sum_{\substack{p, m \geq 1 \\ 0 < m \log p / \log \lambda < 2}} \frac{\log p}{p^{m/2}} D_3\left(\frac{m \log p}{\log \lambda}\right).$$

Split it into

$$D_{\text{tot}}(\lambda) = D_f(\lambda) + D_n(\lambda),$$

where D_f contains the first-order shifts with $p \in [\lambda^{0.7}, \lambda]$ and D_n contains the remaining terms.

Lemma 5.1 (Common frontier Rayleigh vector). *Let $e_1 = (0, 1, 0)^T$. For all $r \in [0.7, 1]$,*

$$e_1^T D_3(r) e_1 \leq -C_f, \quad C_f = 0.4685.$$

Proof. By [Theorem 3.2](#),

$$e_1^T D_3(r) e_1 = (2 - r)(\cos \pi r - \cos 2\pi r) - \frac{\sin \pi r}{\pi} - \frac{\sin 2\pi r}{2\pi}.$$

Interval evaluation of this one-variable trigonometric function on $[0.7, 1]$ gives maximum < -0.4685 , attained at the left endpoint. Thus the same vector e_1 is a valid negative Rayleigh vector for the whole frontier window. $\square$

Lemma 5.2 (Frontier lower bound). *For the frontier block,*

$$\lambda_{\min}(D_f(\lambda)) \leq -C_f W_f(\lambda), \quad W_f(\lambda) = \sum_{\lambda^{0.7} \leq p \leq \lambda} \frac{\log p}{\sqrt{p}}.$$

Furthermore,

$$W_f(\lambda) = 2\sqrt{\lambda}(1 - \lambda^{-0.15}) + O(1).$$

Proof. Use the fixed Rayleigh vector e_1 from [Theorem 5.1](#):

$$\lambda_{\min}(D_f) \leq e_1^T D_f e_1 \leq -C_f \sum_{\lambda^{0.7} \leq p \leq \lambda} \frac{\log p}{\sqrt{p}}.$$

The asymptotic estimate follows by partial summation [\[7\]](#) from the Prime Number Theorem in the form $\theta(x) \sim x$. $\square$

Lemma 5.3 (Non-frontier norm bound). *There is an absolute constant C_n such that*

$$\|D_n(\lambda)\|_{\text{op}} \leq C_n(2\lambda^{0.35} + O(\log \lambda)).$$

One may take $C_n = 2.2847$ from the interval-certified bound $\sup_{0 < r < 2} \|D_3(r)\|_{\text{op}} \leq 2.2847$.

Proof. The first non-frontier part, $p < \lambda^{0.7}$ with $m = 1$, has total weight

$$\sum_{p \leq \lambda^{0.7}} \frac{\log p}{\sqrt{p}} = 2\lambda^{0.35} + O(1)$$

by partial summation with the unconditional estimate $|\theta(x) - x| \leq 0.006788x/\log x$ for $x \geq 89,967,803$ [\[3\]](#); for smaller x the sum is bounded by direct computation. The higher-shift terms satisfy

$$\sum_{p \leq \lambda} \sum_{m \geq 2} \frac{\log p}{p^{m/2}} = O(\log \lambda),$$

with the $m = 2$ term giving the logarithmic growth and $m \geq 3$ bounded by convergent prime sums. Multiplying by the uniform operator-norm bound for $D_3(r)$ gives the claim. $\square$

Proposition 5.4 (Asymptotic frontier dominance). *There exists an explicit Λ_* such that for all $\lambda \geq \Lambda_*$,*

$$\lambda_{\min}(D_{\text{tot}}(\lambda)) < 0.$$

With the constants above, Λ_ lies below the certified endpoint 1,300,000 after the explicit error terms are evaluated.*

Proof. We argue via the Rayleigh quotient at the fixed common test vector $e_1 = (0, 1, 0)^T$ together with the Cauchy–Schwarz operator-norm bound (no eigenvalue additivity is invoked). The variational inequality gives

$$\lambda_{\min}(D_{\text{tot}}) \leq e_1^T D_{\text{tot}} e_1 = e_1^T D_f e_1 + e_1^T D_n e_1.$$

For the frontier block, [Theorems 5.1](#) and [5.2](#) give $e_1^T D_f e_1 \leq -C_f W_f(\lambda) = -2C_f \sqrt{\lambda}(1 - \lambda^{-0.15}) + O(1)$. For the non-frontier block, Cauchy–Schwarz yields $|e_1^T D_n e_1| \leq \|e_1\|^2 \|D_n\|_{\text{op}} = \|D_n\|_{\text{op}}$, which is bounded by [Theorem 5.3](#). Combining,

$$\lambda_{\min}(D_{\text{tot}}) \leq -2C_f \sqrt{\lambda}(1 - \lambda^{-0.15}) + 2C_n \lambda^{0.35} + O(\log \lambda).$$

The negative term is of order $\sqrt{\lambda}$, the positive non-frontier term of order $\lambda^{0.35}$, so the right-hand side is strictly negative for all sufficiently large λ . Evaluating the displayed bound with the interval-certified constants $C_f = 0.4685$, $C_n = 2.2847$ and the explicit PNT/Mertens remainders gives an effective threshold $\Lambda_* \leq 1,300,000$; see [[5](#), Proposition A6 (Direct Frontier-Dominance), Remark on direct-proof ingredients]. The asymptotic regime $\lambda \geq \Lambda_*$ and the certified range $100 \leq \lambda \leq 1,300,000$ of [Theorem 4.3](#) therefore meet without gap. $\square$

Remark 5.5 (Computational improvement of Λ_*). The bound $\Lambda_* \leq 1,300,000$ above is conservative: it is the right endpoint of the certified interval. The exact Rayleigh gap

$$g(\lambda) = -C_f W_f(\lambda) + C_n W_n(\lambda)$$

(with W_f, W_n the precise frontier and non-frontier prime-sum weights, computed by exact prime enumeration) was evaluated as follows.

(i) *Step-1 exhaustive check* [171,329, 182,000]: Every integer λ in this range was checked; $g(\lambda) < 0$ throughout, with the first negative value $g(171,329) = -0.0052$.

(ii) *Step-1000 coarse scan* [182,000, 300,000]: All sampled values give $g \leq -3.5$. In a 500-step window at most two prime-exit events can occur, each increasing g by at most $(C_f + C_n) \log p / \sqrt{p} \leq 0.35$; the total upward perturbation ≤ 0.70 leaves a margin of ≥ 2.8 below zero. No unsampled integer in [182,000, 300,000] can be positive.

(iii) *Asymptotic regime* $\lambda \geq 300,000$: $g \leq -43$ throughout (coarse scan, step 10,000), consistent with the $-\Theta(\sqrt{\lambda})$ dominance of the asymptotic formula.

Combining (i)–(iii) with the certified range [100, 1,300,000] gives $\Lambda_* \leq 175,000$. The verification script `verify_lambda_star.py` (float64, safety margins $\gg 10^{-10}$) is archived in the companion repository at <https://github.com/research-line/rh-even-dominance>.

6 Tail Control and Lifting to the Weil Form

[Theorem 5.4](#) establishes $\lambda_{\min}(D_{\text{tot}}(\lambda)) < 0$ in the three-mode difference block. The full Weil operator A_λ acts on $L^2([-L, L])$ with infinitely many Fourier modes and includes the archimedean integral. We show that the contributions beyond the low-mode prime block preserve the sign of the even–odd gap.

Remark 6.1 (Archimedean contribution). The archimedean kernel $e^{u/2}/(2 \sinh u)$ in (1) is integrable on $(0, \infty)$, and the sector-independent term $-2e^{-u/2} \|f\|^2$ cancels in the even-odd difference. The remaining archimedean contribution involves shift overlap differences $D_N(u/L)$ weighted against the fixed kernel. Since $\|D_3(r)\|_{\text{op}} \leq 2.2847$ uniformly and the kernel integral is a constant independent of λ , the archimedean even-odd contribution satisfies $|\Delta_{\text{arch}}| \leq C_{\text{arch}}$ for an absolute constant. As the prime contribution is $\Theta(\sqrt{\lambda})$ by [Theorem 5.2](#), the archimedean part is negligible for large λ .

Definition 6.2 (Auto-overlap profile). *For the even basis function φ_n and normalized shift $u = \delta/L \in [0, 2]$, define $h_n(u) := \frac{1}{4} \langle \varphi_n, (S_{uL} + S_{-uL}) \varphi_n \rangle$. The overlap integral evaluates via the substitution $x = t/L$ and the product-to-sum identity $\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$:*

$$\begin{aligned} \langle \varphi_n, (S_{uL} + S_{-uL}) \varphi_n \rangle &= 2 \int_{u-1}^1 \cos(n\pi x) \cos(n\pi(x-u)) dx \\ &= \cos(n\pi u)(2-u) + \frac{\sin(n\pi(2-u)) - \sin(n\pi(u-2))}{2n\pi} \\ &= (2-u) \cos(n\pi u) - \frac{\sin(n\pi u)}{n\pi}, \end{aligned}$$

where the last step uses $\sin(n\pi(2-u)) = -\sin(n\pi u)$ for integer n . Dividing by 4:

$$h_n(u) = \frac{1}{2} \left(1 - \frac{u}{2}\right) \cos(n\pi u) - \frac{\sin(n\pi u)}{4n\pi}, \quad n \geq 1,$$

with $h_0(u) = (1 - u/2)/2$ (from $\langle \varphi_0, (S_{uL} + S_{-uL}) \varphi_0 \rangle = 2 - u$ by direct overlap-length computation). At the Laplace endpoint $u = 1$: $h_n(1) = (-1)^n/4$ for all $n \geq 0$. The normalization factor $\frac{1}{4}$ is chosen so that the prime-weighted diagonal satisfies $(W_{\text{prime}})_{nn} = \sum_p (\log p) p^{-1/2} \cdot 4h_n(\log p/L)$, consistent with the Weil form (1).

Lemma 6.3 (Leading-mode cancellation). *Define the symmetrized off-diagonal overlap profiles $F_j(u) := S_{\text{sym}}^-(0, j; u)$ (odd sector) and $G_j(u) := S_{\text{sym}}^+(1, j'; u)$ (even sector) for the leading modes. Their closed forms are:*

$$S_{\text{sym}}^+(1, 0; u) = \frac{\sqrt{2} \sin(\pi u)}{\pi}, \quad (2)$$

$$S_{\text{sym}}^+(1, 2; u) = \frac{2(4 \cos \pi u - 1) \sin \pi u}{3\pi}, \quad (3)$$

$$S_{\text{sym}}^-(0, 1; u) = \frac{2(-2 \sin \pi u + \sin 2\pi u)}{3\pi}, \quad (4)$$

$$S_{\text{sym}}^-(0, 2; u) = \frac{3 \sin \pi u - \sin 3\pi u}{4\pi}. \quad (5)$$

The overlap differences $\Delta_j(u) := F_j(u) - G_j(u)$ satisfy:

- (a) For the first two energy slots, the individual differences are $O(1)$, but their sum satisfies $\Delta_{\text{slot } 1}(u) + \Delta_{\text{slot } 2}(u) = -(2 + \sqrt{2})u + O(u^2)$.
- (b) For $j \geq 2$: $\Delta_j(u) = O(u)$ individually.

Proof. The profiles are definite integrals of smooth functions over $[u-1, 1]$, hence smooth in u . For example, $S_{\text{sym}}^+(1, 0; u) = \langle \varphi_1, \varphi_0(\cdot - uL) + \varphi_0(\cdot + uL) \rangle$ evaluates to $\frac{1}{L\sqrt{2}} \int_{(u-1)L}^L \cos(\pi t/L) dt + (\text{symmetric}) = \frac{\sqrt{2} \sin \pi u}{\pi}$ by direct integration; formulas (3)–(5) are obtained similarly. At $u = 0$, Fourier orthogonality gives $F_j(0) = G_j(0) = 0$. Differentiating (4)–(5) at $u = 0$: both sine overlaps have zero derivative (Taylor expansions start at u^2). For (4): rewrite as $-2 \sin \pi u + \sin 2\pi u = 2 \sin \pi u (\cos \pi u - 1)$, vanishing to second order. For (5): rewrite as $3 \sin \pi u - \sin 3\pi u = 4 \sin^3 \pi u$, vanishing to third order. The cosine derivatives are $G'_0(0) = \sqrt{2}$ from (2) and $G'_2(0) = 2$ from (3). Therefore $\sum_{\text{slots}} \Delta'_j(0) = (0 - \sqrt{2}) + (0 - 2) = -(2 + \sqrt{2})$. Part (b) follows from the mean value theorem applied to $\Delta_j(u) = \Delta_j(u) - \Delta_j(0)$. $\square$

Lemma 6.4 (Higher-mode decay). *Let $B_{1j}^\pm$ denote the coupling between the leading even/odd mode and mode $j \geq 2$ induced by the prime shifts, and let g_j denote the spectral gap between mode j and the leading mode in the Galerkin diagonal. Then:*

- (i) Coupling bound: $|B_{1j}^\pm| \leq 4\pi\sqrt{\lambda}/L \cdot (1 + O(1/L))$.
- (ii) Gap bound: $g_j \geq 4\pi^2(j^2 - 1)\sqrt{\lambda}/L^2$ for all $j \geq 2$ and $L \geq 5$.
- (iii) Sector difference: $|\Delta_{\text{high}}|/D(\lambda) = O(1/L) \rightarrow 0$, where $D(\lambda) = \Theta(\sqrt{\lambda})$ is the three-mode frontier advantage.

Proof. (i) *Coupling bound.* For $j \neq 1$, orthogonality of the cosine basis on $[-L, L]$ gives $\langle \varphi_1, (S_0 + S_0)\varphi_j \rangle = 0$. The mean-value theorem yields $|\langle \varphi_1, (S_\delta + S_{-\delta})\varphi_j \rangle| \leq \pi\delta/L$. Summing over primes with Abel summation [7] and $\theta(x) = x + O(xe^{-c\sqrt{\log x}})$:

$$|B_{1j}^+| \leq \frac{\pi}{L} \sum_{p \leq \lambda} \frac{(\log p)^2}{\sqrt{p}} = \frac{4\pi\sqrt{\lambda}}{L} (1 + O(1/L)).$$

The same bound holds for $|B_{0j}^-|$ in the odd sector.

(ii) *Gap bound.* By Theorem 6.2, $h_n(1) = (-1)^n/4$. The prime sum concentrates at $p \sim \lambda$ (i.e., $u = \log p / \log \lambda \rightarrow 1$); by Abel summation with the PNT [7], the diagonal satisfies $(W_{\text{prime}})_{nn} \sim 4h_n(1)\sqrt{\lambda} = (-1)^n\sqrt{\lambda}$. For even j : $h_j(1) - h_1(1) = \frac{1}{2}$, giving $g_j \geq \sqrt{\lambda}$. For odd $j \geq 3$: $h_j(1) - h_1(1) = -\frac{1}{4}$ (leading terms cancel). Expanding $h_j(u) - h_1(u)$ about $u = 1$ to second order gives $h_j''(1) - h_1''(1) = \pi^2(j^2 - 1)/4$, whence $g_j = 4\pi^2(j^2 - 1)\sqrt{\lambda}/L^2 + O(\sqrt{\lambda}/L^3)$. In particular, $c_{\text{gap}} := 4\pi^2/L \geq 2.8$ for $L \leq 14$ ($\lambda \leq 1,200,000$).

(iii) *Sector difference.* Let $E_{\text{cos}} := \sum_{j \geq 2} |B_{1j}^+|^2/g_j^+$ and $E_{\text{sin}} := \sum_{j \geq 2} |B_{0j}^-|^2/g_j^-$ denote the resolvent-damped coupling energies in the even and odd sectors. Each is $O(\sqrt{\lambda})$ individually by (i)–(ii). The sector difference $|E_{\text{sin}} - E_{\text{cos}}|$ is further suppressed by the Leading-Mode Cancellation (Theorem 6.3): the first-order Taylor coefficients of the even and odd overlap differences at $u = 0$ sum to $-(2 + \sqrt{2})$, so the leading contributions cancel pairwise. The coefficients $c_{\text{sin}}(L) - c_{\text{cos}}(L) = O(1/L^2)$, and therefore $|\Delta_{\text{high}}|/D(\lambda) = O(1/L) \rightarrow 0$. $\square$

Lemma 6.5 (Galerkin truncation error). *The even sector is dominated by the frontier mode ($n = 1$, diagonal $\sim -\sqrt{\lambda}$); modes $n = 0, 2, 3$ contribute at lower order, so $N_{\text{cos}} = 4$ suffices (Theorem 4.1). The odd sector lacks this structural amplification and is enclosed with $N_0 = N_{\text{sin}} \in [40, 90]$. By the Schur-complement formula [8], the eigenvalue perturbation from modes $j > N_0$ is bounded by*

$$|\lambda_1^\pm(\text{full}) - \lambda_1^\pm(N_0)| \leq \frac{\|B_{\text{tail}}\|_{\text{op}}^2}{g_{N_0+1} - \|B_{\text{tail}}\|_{\text{op}}}.$$

Substituting Theorem 6.4(i)–(ii) gives $\|B_{\text{tail}}\|_{\text{op}} \leq 8\pi\sqrt{\lambda}/L$ and $g_{N_0+1} \geq 4\pi^2 N_0^2 \sqrt{\lambda}/L^2$, so

$$|\lambda_1^\pm(\text{full}) - \lambda_1^\pm(N_0)| \leq \frac{16\sqrt{\lambda}}{N_0^2} (1 + o(1)) \quad \text{as } \lambda \rightarrow \infty.$$

At every certified grid point this analytical bound is dominated by the interval-certified Landscape safety factor of Theorem 4.2 ($\geq 34\times$ at $\lambda = 100$, $\geq 65\times$ for $\lambda \geq 200$), which is the rigorous statement used in the certified gap.

Proof. The Schur-complement bound is standard [8]. Theorem 6.4(i) gives $\|B_{\text{tail}}\|_{\text{op}} \leq 8\pi\sqrt{\lambda}/L$. Theorem 6.4(ii) gives $g_{N_0+1} \geq 4\pi^2 N_0^2 \sqrt{\lambda}/L^2 \gg \|B_{\text{tail}}\|_{\text{op}}$ for $N_0 \geq 40$, hence the denominator is dominated by g_{N_0+1} . The displayed bound $16\sqrt{\lambda}/N_0^2$ follows. The certified gap uses the sharper interval-arithmetic enclosure of [5, Section 2.9] with the Cauchy tail bound described in Theorem 4.2. $\square$

7 Main Theorem

Theorem 7.1 (Even dominance for a divergent sequence). *There exists a sequence $\lambda_n \rightarrow \infty$ such that QW_{λ_n} has a simple even ground state for every n .*

Proof. For $100 \leq \lambda \leq 1,300,000$, the certified grid of [Theorem 4.3](#) gives even dominance and even-sector simplicity at the listed values; these certificates evaluate the full Galerkin operator (including archimedean and higher-mode contributions).

For λ beyond the asymptotic threshold, [Theorem 5.4](#) gives a three-mode gap $\Delta_{\text{low}} = -\Theta(\sqrt{\lambda})$. The higher-mode correction satisfies $|\Delta_{\text{high}}| = O(\sqrt{\lambda}/L) = o(|\Delta_{\text{low}}|)$ by [Theorem 6.4\(iii\)](#), the archimedean contribution is $O(1)$ by [Theorem 6.1](#), and the Galerkin truncation error is $\leq 16\sqrt{\lambda}/N_0^2$ by [Theorem 6.5](#) – a fraction $4/N_0^2$ of the leading $\sqrt{\lambda}$ -gap, hence $o(|\Delta_{\text{low}}|)$ for $N_0 \geq 40$. Therefore the full even–odd gap is negative for all sufficiently large λ , with the asymptotic threshold meeting the certified range exactly at $\Lambda_* \leq 175,000$ (see [Theorem 5.5](#); the conservative upper bound $\Lambda_* \leq 1,300,000$ of [Theorem 5.4](#) is also valid). Taking any divergent sequence of certified or asymptotic values gives even dominance.

Simplicity of the even ground state follows from the intra-even spectral gap $\lambda_2^+ - \lambda_1^+ = \Theta(\sqrt{\lambda})$: at the certificate points this is directly verified ([Theorem 4.3](#)); for large λ it follows from the Galerkin diagonal asymptotics, since the leading even mode ($n = 1$) has $(W_{\text{prime}})_{11} \sim -\sqrt{\lambda}$ while the next mode ($n = 0$) has a different Laplace endpoint value. $\square$

Theorem 7.2 (Riemann Hypothesis, conditional on the Connes program). *Conditional on [1, Theorem 6.1 and the Hurwitz bridge of Section 6.4–6.6] and [2], all non-trivial zeros of the Riemann zeta function [9] lie on the line $\Re(s) = 1/2$.*

Proof. Apply [Theorem 2.3](#) to the sequence supplied by [Theorem 7.1](#). The Fourier transforms associated with the even-dominant Weil forms have only real zeros by [Theorem 2.2](#). Hurwitz transfer gives the real-zero property for the limiting Ξ -function. Therefore the corresponding non-trivial zeros of $\zeta(s)$ lie on the critical line. $\square$

8 What Is Not Used

The proof does not use the following Landscape material:

- the original gauge-thermodynamic route and its obstructions;
- an absolute PF_∞ or total-positivity claim for the full prime operator;
- a universal commuting Sturm–Liouville operator;
- Hellmann–Feynman monotonicity between certificate points;
- the narrative Dungeon/Memorial presentation of excluded routes.

Those results remain valuable as route classification and audit history, but they are not part of the proof chain above.

A Constants Audit Table

All explicit numerical constants used in the proof, with their source, role, and verification mechanism. Constants flagged “IA” are interval-arithmetic certified (`mpmath.iv`, 50-digit precision); “CAP” are the 33 Landscape computer-assisted certificates; “PNT/Mertens” are explicit asymptotic remainders.

Constant	Value	Role / Source	Verification
C_f	0.4685	Common frontier Rayleigh quotient $-d_{11}(r) \geq C_f$ on $r \in [0.7, 1]$ (Theorem 5.1)	IA on $[0.7, 1]$ (1D)
C_n	2.2847	Operator-norm bound $\sup_{r \in (0,2)} \ D_3(r)\ _{\text{op}} \leq C_n$ (Theorem 5.3)	IA on $(0, 2)$ (3D)
c_{gap}	$\geq 4\pi^2/L \geq 2.8$	Spectral-gap constant for $L \leq 14$ (Theorem 6.4(ii))	Laplace asymptotics
N_{cos}	4	Even-block Galerkin truncation (Theorem 4.1)	Stability check $k = 3, 4, 5$
N_{sin}	40–90	Odd-block Galerkin truncation (Theorem 4.1)	Cauchy tail bound
Λ_*	$\leq 175,000$	Asymptotic threshold; prime-sum computation (Theorem 5.5); conservative bound $\leq 1,300,000$ (Theorem 5.4)	Exact prime enum.
$\theta(x) - x$	$\leq 0.006788 \frac{x}{\log x}$	Effective Chebyshev bound for $x \geq 89,967,803$ (Dusart 2010, [3])	Unconditional
$ \text{cert_gap} $ at $\lambda=100$	≥ 2.25	First CAP (Theorem 4.3)	CAP
$ \text{cert_gap} $ at $\lambda=1.3 \cdot 10^6$	≥ 475.83	Last CAP	CAP
Safety factor at $\lambda=100$	$\geq 34\times$	$ \text{cert_gap} $ vs. Galerkin/tail error (Theorem 4.2)	IA + Cauchy tail
Safety factor at $\lambda \geq 200$	$\geq 65\times$	Same, monotone in λ	IA + Cauchy tail
c (slot cancellation)	$2 + \sqrt{2}$	Closed-form constant (Theorem 6.3(a))	Symbolic (sympy)

The Landscape companion repository archives the verification scripts. The most relevant entries are:

- `shift_parity_cert_v3_targeted.py` – Shift Parity Lemma;
- `certifier_production.py` – the 33 CAP certificates;
- `partA_proof_sketch.py` – frontier-dominance check;
- `euler_maclaurin_certifier.py` – asymptotic threshold;

see [5, scripts/]. A dedicated companion repository for the present extraction is mirrored at <https://github.com/research-line/rh-even-dominance>.

AI Disclosure

Statement on the Use of AI Tools

This manuscript was prepared with moderate assistance from AI-based language models (Claude, Anthropic; Gemini, Google; ChatGPT, OpenAI). The use encompassed the following areas:

- *Research*: Assistance with systematic literature search, identification and summarisation of relevant research literature.
- *Structuring*: Support in outlining and logically organising the manuscript, in particular the proof-only extraction from the Landscape series.
- *Drafting*: Assistance in formulating individual sections and paragraphs based on core arguments and theses specified by the author.

- *Language*: Grammar and spell checking, linguistic revision.
- *Formatting*: Preparation and verification of citations and bibliographies; L^AT_EX typesetting support.
- *Internal review*: A 7-phase model-based review chain (constructive, expert, adversarial) was used to identify and address structural and citation issues prior to submission.

The entire research process – including the research question, methodological decisions, substantive argumentation, critical evaluation of sources, and conclusions – was conceived, directed, and borne by the author. All computer-assisted certificates were computed by deterministic numerical scripts independently of AI. All AI-generated suggestions were critically reviewed, revised, and where appropriate rejected. The author reviewed all content to the best of his knowledge but cannot guarantee complete independent verification of all factual claims and references.

Full scholarly and substantive responsibility for this article rests with the author, Lukas Geiger.

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